

```

In [ ]: Enter password: ****
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 39
Server version: 8.0.44 MySQL Community Server - GPL

Copyright (c) 2000, 2025, Oracle and/or its affiliates.

Oracle is a registered trademark of Oracle Corporation and/or its
affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> show databases;
+-----+
| Database |
+-----+
| customers |
| information_schema |
| mysql |
| nit |
| performance_schema |
| sys |
| world |
+-----+
7 rows in set (0.03 sec)

mysql> use nit;
Database changed
mysql> show tables;
+-----+
| Tables_in_nit |
+-----+
| student |
+-----+
1 row in set (0.01 sec)

mysql> select * from student;
+-----+-----+-----+-----+
| name | id | address | marks |
+-----+-----+-----+-----+
| prakash | 12 | hyd | 78 |
| rakesh | 18 | delhi | 87 |
| kodi | 40 | chennai | 66 |
| likhith | 45 | hyd | 90 |
| srikanth | 48 | pune | 70 |
| raj | 78 | mumbai | 88 |
+-----+-----+-----+-----+
6 rows in set (0.01 sec)

mysql> alter table student add phoneNo int;
Query OK, 0 rows affected (0.04 sec)
Records: 0 Duplicates: 0 Warnings: 0

```

```
mysql> select * from student;
```

| name     | id | address | marks | phoneNo |
|----------|----|---------|-------|---------|
| prakash  | 12 | hyd     | 78    | NULL    |
| rakesh   | 18 | delhi   | 87    | NULL    |
| kodi     | 40 | chennai | 66    | NULL    |
| likhith  | 45 | hyd     | 90    | NULL    |
| srikanth | 48 | pune    | 70    | NULL    |
| raj      | 78 | mumbai  | 88    | NULL    |

```
6 rows in set (0.00 sec)
```

```
mysql> update student set phoneNo=123;
```

```
Query OK, 6 rows affected (0.01 sec)
```

```
Rows matched: 6 Changed: 6 Warnings: 0
```

```
mysql> select * from student;
```

| name     | id | address | marks | phoneNo |
|----------|----|---------|-------|---------|
| prakash  | 12 | hyd     | 78    | 123     |
| rakesh   | 18 | delhi   | 87    | 123     |
| kodi     | 40 | chennai | 66    | 123     |
| likhith  | 45 | hyd     | 90    | 123     |
| srikanth | 48 | pune    | 70    | 123     |
| raj      | 78 | mumbai  | 88    | 123     |

```
6 rows in set (0.00 sec)
```

```
mysql> update student set phoneNo=456 where id=12;
```

```
Query OK, 1 row affected (0.01 sec)
```

```
Rows matched: 1 Changed: 1 Warnings: 0
```

```
mysql> select * from student;
```

| name     | id | address | marks | phoneNo |
|----------|----|---------|-------|---------|
| prakash  | 12 | hyd     | 78    | 456     |
| rakesh   | 18 | delhi   | 87    | 123     |
| kodi     | 40 | chennai | 66    | 123     |
| likhith  | 45 | hyd     | 90    | 123     |
| srikanth | 48 | pune    | 70    | 123     |
| raj      | 78 | mumbai  | 88    | 123     |

```
6 rows in set (0.00 sec)
```

```
mysql> desc student;
```

| Field   | Type        | Null | Key | Default | Extra |
|---------|-------------|------|-----|---------|-------|
| name    | varchar(30) | YES  |     | NULL    |       |
| id      | int         | NO   | PRI | NULL    |       |
| address | varchar(50) | YES  |     | NULL    |       |
| marks   | int         | YES  |     | NULL    |       |
| phoneNo | int         | YES  |     | NULL    |       |

```

+-----+-----+-----+-----+-----+
5 rows in set (0.01 sec)

mysql> alter table student modify column name varchar(20);
Query OK, 6 rows affected (0.09 sec)
Records: 6 Duplicates: 0 Warnings: 0

mysql> desc student;
+-----+-----+-----+-----+-----+-----+
| Field | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| name  | varchar(20)   | YES  |     | NULL    |       |
| id    | int           | NO   | PRI | NULL    |       |
| address | varchar(50)  | YES  |     | NULL    |       |
| marks | int           | YES  |     | NULL    |       |
| phoneNo | int          | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)

mysql> alter table student drop column phoneNo;
Query OK, 0 rows affected (0.03 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> select * from student;
+-----+-----+-----+-----+
| name  | id | address | marks |
+-----+-----+-----+-----+
| prakash | 12 | hyd     | 78    |
| rakesh  | 18 | delhi   | 87    |
| kodi    | 40 | chennai | 66    |
| likhith | 45 | hyd     | 90    |
| srikanth | 48 | pune    | 70    |
| raj     | 78 | mumbai  | 88    |
+-----+-----+-----+-----+
6 rows in set (0.00 sec)

mysql> delete from student where name=kodi;
ERROR 1054 (42S22): Unknown column 'kodi' in 'where clause'
mysql> delete from student where name='kodi';
Query OK, 1 row affected (0.01 sec)

mysql> select * from student;
+-----+-----+-----+-----+
| name  | id | address | marks |
+-----+-----+-----+-----+
| prakash | 12 | hyd     | 78    |
| rakesh  | 18 | delhi   | 87    |
| likhith | 45 | hyd     | 90    |
| srikanth | 48 | pune    | 70    |
| raj     | 78 | mumbai  | 88    |
+-----+-----+-----+-----+
5 rows in set (0.00 sec)

mysql> select sum(marks) from student;
+-----+
| sum(marks) |
+-----+

```

```

+-----+
|      413 |
+-----+
1 row in set (0.01 sec)

mysql> select avg(marks) from student;
+-----+
| avg(marks) |
+-----+
|      82.6000 |
+-----+
1 row in set (0.00 sec)

mysql> select count(name) from student;
+-----+
| count(name) |
+-----+
|           5 |
+-----+
1 row in set (0.04 sec)

mysql> select max(marks) from student;
+-----+
| max(marks) |
+-----+
|          90 |
+-----+
1 row in set (0.00 sec)

mysql> select min(marks) from student;
+-----+
| min(marks) |
+-----+
|          70 |
+-----+
1 row in set (0.00 sec)

mysql> select * from student;
+-----+-----+-----+-----+
| name      | id  | address | marks |
+-----+-----+-----+-----+
| prakash   | 12  | hyd     | 78    |
| rakesh    | 18  | delhi   | 87    |
| likhith   | 45  | hyd     | 90    |
| srikanth  | 48  | pune    | 70    |
| raj       | 78  | mumbai  | 88    |
+-----+-----+-----+-----+
5 rows in set (0.00 sec)

mysql> select * from student order by marks;
+-----+-----+-----+-----+
| name      | id  | address | marks |
+-----+-----+-----+-----+
| srikanth  | 48  | pune    | 70    |
| prakash   | 12  | hyd     | 78    |
| rakesh    | 18  | delhi   | 87    |

```

|         |    |        |    |
|---------|----|--------|----|
| raj     | 78 | mumbai | 88 |
| likhith | 45 | hyd    | 90 |

```
+-----+-----+-----+-----+
```

5 rows in set (0.00 sec)

```
mysql> select * from student order by marks desc;
```

| name | id | address | marks |
|------|----|---------|-------|
|------|----|---------|-------|

```
+-----+-----+-----+-----+
```

|          |    |        |    |
|----------|----|--------|----|
| likhith  | 45 | hyd    | 90 |
| raj      | 78 | mumbai | 88 |
| rakesh   | 18 | delhi  | 87 |
| prakash  | 12 | hyd    | 78 |
| srikanth | 48 | pune   | 70 |

```
+-----+-----+-----+-----+
```

5 rows in set (0.00 sec)

```
mysql> select * from student where name like 'r%';
```

| name | id | address | marks |
|------|----|---------|-------|
|------|----|---------|-------|

```
+-----+-----+-----+-----+
```

|        |    |        |    |
|--------|----|--------|----|
| rakesh | 18 | delhi  | 87 |
| raj    | 78 | mumbai | 88 |

```
+-----+-----+-----+-----+
```

2 rows in set (0.00 sec)

```
mysql> select * from student where name like '%j';
```

| name | id | address | marks |
|------|----|---------|-------|
|------|----|---------|-------|

```
+-----+-----+-----+-----+
```

|     |    |        |    |
|-----|----|--------|----|
| raj | 78 | mumbai | 88 |
|-----|----|--------|----|

```
+-----+-----+-----+-----+
```

1 row in set (0.00 sec)

```
mysql> select * from student where name like '_i%';
```

| name | id | address | marks |
|------|----|---------|-------|
|------|----|---------|-------|

```
+-----+-----+-----+-----+
```

|         |    |     |    |
|---------|----|-----|----|
| likhith | 45 | hyd | 90 |
|---------|----|-----|----|

```
+-----+-----+-----+-----+
```

1 row in set (0.00 sec)

```
mysql> select * from student where name like '_r%';
```

| name | id | address | marks |
|------|----|---------|-------|
|------|----|---------|-------|

```
+-----+-----+-----+-----+
```

|          |    |      |    |
|----------|----|------|----|
| prakash  | 12 | hyd  | 78 |
| srikanth | 48 | pune | 70 |

```
+-----+-----+-----+-----+
```

2 rows in set (0.00 sec)

```
mysql> select * from student where name like '_k%';
```

Empty set (0.00 sec)

```
mysql> create table emp(id int not null primary key,salary int,empcode int,name var
Query OK, 0 rows affected (0.03 sec)
```

```
mysql> insert into emp values(12,20000,102,'aman'),(22,25000,104,'ram'),(39,,34000,
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that cor
mysql> insert into emp values(12,20000,102,'aman'),(22,25000,104,'ram'),(39,,34000,
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that cor
mysql> insert into emp values(12,20000,102,'aman'),(22,25000,104,'ram'),(39,34000,1
Query OK, 5 rows affected (0.01 sec)
Records: 5 Duplicates: 0 Warnings: 0
```

```
mysql> select * from emp;
```

| id | salary | empcode | name  |
|----|--------|---------|-------|
| 12 | 20000  | 102     | aman  |
| 22 | 25000  | 104     | ram   |
| 28 | 62500  | 101     | srinu |
| 30 | 50000  | 105     | sam   |
| 39 | 34000  | 107     | kumar |

5 rows in set (0.00 sec)

```
mysql> show tables;
```

|               |
|---------------|
| Tables_in_nit |
| emp           |
| student       |

2 rows in set (0.00 sec)

```
mysql> select * from student;
```

| name     | id | address | marks |
|----------|----|---------|-------|
| prakash  | 12 | hyd     | 78    |
| rakesh   | 18 | delhi   | 87    |
| likhith  | 45 | hyd     | 90    |
| srikanth | 48 | pune    | 70    |
| raj      | 78 | mumbai  | 88    |

5 rows in set (0.00 sec)

```
mysql> select * from student inner join emp on student.id=emp.id;
```

| name    | id | address | marks | id | salary | empcode | name |
|---------|----|---------|-------|----|--------|---------|------|
| prakash | 12 | hyd     | 78    | 12 | 20000  | 102     | aman |

1 row in set (0.00 sec)

```
mysql> select * from emp inner join student on emp.id=student.id;
```

| id | salary | empcode | name | name    | id | address | marks |
|----|--------|---------|------|---------|----|---------|-------|
| 12 | 20000  | 102     | aman | prakash | 12 | hyd     | 78    |

```
1 row in set (0.00 sec)
```

```
mysql> select * from student;
```

| name     | id | address | marks |
|----------|----|---------|-------|
| prakash  | 12 | hyd     | 78    |
| rakesh   | 18 | delhi   | 87    |
| likhith  | 45 | hyd     | 90    |
| srikanth | 48 | pune    | 70    |
| raj      | 78 | mumbai  | 88    |

```
5 rows in set (0.00 sec)
```

```
mysql> select * from emp;
```

| id | salary | empcode | name  |
|----|--------|---------|-------|
| 12 | 20000  | 102     | aman  |
| 22 | 25000  | 104     | ram   |
| 28 | 62500  | 101     | srinu |
| 30 | 50000  | 105     | sam   |
| 39 | 34000  | 107     | kumar |

```
5 rows in set (0.00 sec)
```

```
mysql> select * from student left join emp student.id=emp.id;
```

ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that covers the correct syntax for the statement

```
mysql> select * from student left join emp on student.id=emp.id;
```

| name     | id | address | marks | id   | salary | empcode | name |
|----------|----|---------|-------|------|--------|---------|------|
| prakash  | 12 | hyd     | 78    | 12   | 20000  | 102     | aman |
| rakesh   | 18 | delhi   | 87    | NULL | NULL   | NULL    | NULL |
| likhith  | 45 | hyd     | 90    | NULL | NULL   | NULL    | NULL |
| srikanth | 48 | pune    | 70    | NULL | NULL   | NULL    | NULL |
| raj      | 78 | mumbai  | 88    | NULL | NULL   | NULL    | NULL |

```
5 rows in set (0.00 sec)
```

```
mysql> select * from emp left join student on emp.id=student.id;
```

| id | salary | empcode | name  | name    | id   | address | marks |
|----|--------|---------|-------|---------|------|---------|-------|
| 12 | 20000  | 102     | aman  | prakash | 12   | hyd     | 78    |
| 22 | 25000  | 104     | ram   | NULL    | NULL | NULL    | NULL  |
| 28 | 62500  | 101     | srinu | NULL    | NULL | NULL    | NULL  |
| 30 | 50000  | 105     | sam   | NULL    | NULL | NULL    | NULL  |
| 39 | 34000  | 107     | kumar | NULL    | NULL | NULL    | NULL  |

```
5 rows in set (0.00 sec)
```

```
mysql> select * from emp right join student on emp.id=student.id;
```

| id | salary | empcode | name  | name    | id   | address | marks |
|----|--------|---------|-------|---------|------|---------|-------|
| 12 | 20000  | 102     | aman  | prakash | 12   | hyd     | 78    |
| 22 | 25000  | 104     | ram   | NULL    | NULL | NULL    | NULL  |
| 28 | 62500  | 101     | srinu | NULL    | NULL | NULL    | NULL  |
| 30 | 50000  | 105     | sam   | NULL    | NULL | NULL    | NULL  |
| 39 | 34000  | 107     | kumar | NULL    | NULL | NULL    | NULL  |

|      |       |      |      |          |    |        |    |
|------|-------|------|------|----------|----|--------|----|
| 12   | 20000 | 102  | aman | prakash  | 12 | hyd    | 78 |
| NULL | NULL  | NULL | NULL | rakesh   | 18 | delhi  | 87 |
| NULL | NULL  | NULL | NULL | likhith  | 45 | hyd    | 90 |
| NULL | NULL  | NULL | NULL | srikanth | 48 | pune   | 70 |
| NULL | NULL  | NULL | NULL | raj      | 78 | mumbai | 88 |

5 rows in set (0.00 sec)

```
mysql> select * from student rigght join emp on student.id=emp.id;
ERROR 1054 (42S22): Unknown column 'student.id' in 'on clause'
mysql> select * from student right join emp on student.id=emp.id;
```

| name    | id   | address | marks | id | salary | empcode | name  |
|---------|------|---------|-------|----|--------|---------|-------|
| prakash | 12   | hyd     | 78    | 12 | 20000  | 102     | aman  |
| NULL    | NULL | NULL    | NULL  | 22 | 25000  | 104     | ram   |
| NULL    | NULL | NULL    | NULL  | 28 | 62500  | 101     | srinu |
| NULL    | NULL | NULL    | NULL  | 30 | 50000  | 105     | sam   |
| NULL    | NULL | NULL    | NULL  | 39 | 34000  | 107     | kumar |

5 rows in set (0.00 sec)

```
mysql> select * from student cross join emp;
```

| name     | id | address | marks | id | salary | empcode | name  |
|----------|----|---------|-------|----|--------|---------|-------|
| raj      | 78 | mumbai  | 88    | 12 | 20000  | 102     | aman  |
| srikanth | 48 | pune    | 70    | 12 | 20000  | 102     | aman  |
| likhith  | 45 | hyd     | 90    | 12 | 20000  | 102     | aman  |
| rakesh   | 18 | delhi   | 87    | 12 | 20000  | 102     | aman  |
| prakash  | 12 | hyd     | 78    | 12 | 20000  | 102     | aman  |
| raj      | 78 | mumbai  | 88    | 22 | 25000  | 104     | ram   |
| srikanth | 48 | pune    | 70    | 22 | 25000  | 104     | ram   |
| likhith  | 45 | hyd     | 90    | 22 | 25000  | 104     | ram   |
| rakesh   | 18 | delhi   | 87    | 22 | 25000  | 104     | ram   |
| prakash  | 12 | hyd     | 78    | 22 | 25000  | 104     | ram   |
| raj      | 78 | mumbai  | 88    | 28 | 62500  | 101     | srinu |
| srikanth | 48 | pune    | 70    | 28 | 62500  | 101     | srinu |
| likhith  | 45 | hyd     | 90    | 28 | 62500  | 101     | srinu |
| rakesh   | 18 | delhi   | 87    | 28 | 62500  | 101     | srinu |
| prakash  | 12 | hyd     | 78    | 28 | 62500  | 101     | srinu |
| raj      | 78 | mumbai  | 88    | 30 | 50000  | 105     | sam   |
| srikanth | 48 | pune    | 70    | 30 | 50000  | 105     | sam   |
| likhith  | 45 | hyd     | 90    | 30 | 50000  | 105     | sam   |
| rakesh   | 18 | delhi   | 87    | 30 | 50000  | 105     | sam   |
| prakash  | 12 | hyd     | 78    | 30 | 50000  | 105     | sam   |
| raj      | 78 | mumbai  | 88    | 39 | 34000  | 107     | kumar |
| srikanth | 48 | pune    | 70    | 39 | 34000  | 107     | kumar |
| likhith  | 45 | hyd     | 90    | 39 | 34000  | 107     | kumar |
| rakesh   | 18 | delhi   | 87    | 39 | 34000  | 107     | kumar |
| prakash  | 12 | hyd     | 78    | 39 | 34000  | 107     | kumar |

25 rows in set (0.00 sec)

```
mysql> select * from student inner jion emp on student.id=emp.id;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that cor
```



```
mysql> select * from student inner jion emp on student.id = emp.id;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that cor
mysql> select * from student inner join emp on student.id = emp.id;
+-----+-----+-----+-----+-----+-----+-----+-----+
| name   | id | address | marks | id | salary | empcode | name |
+-----+-----+-----+-----+-----+-----+-----+-----+
| prakash | 12 | hyd     | 78    | 12 | 20000  | 102     | aman |
+-----+-----+-----+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)

mysql>
```